# SMART INDIA HACKATHON 2026

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## TITLE PAGE

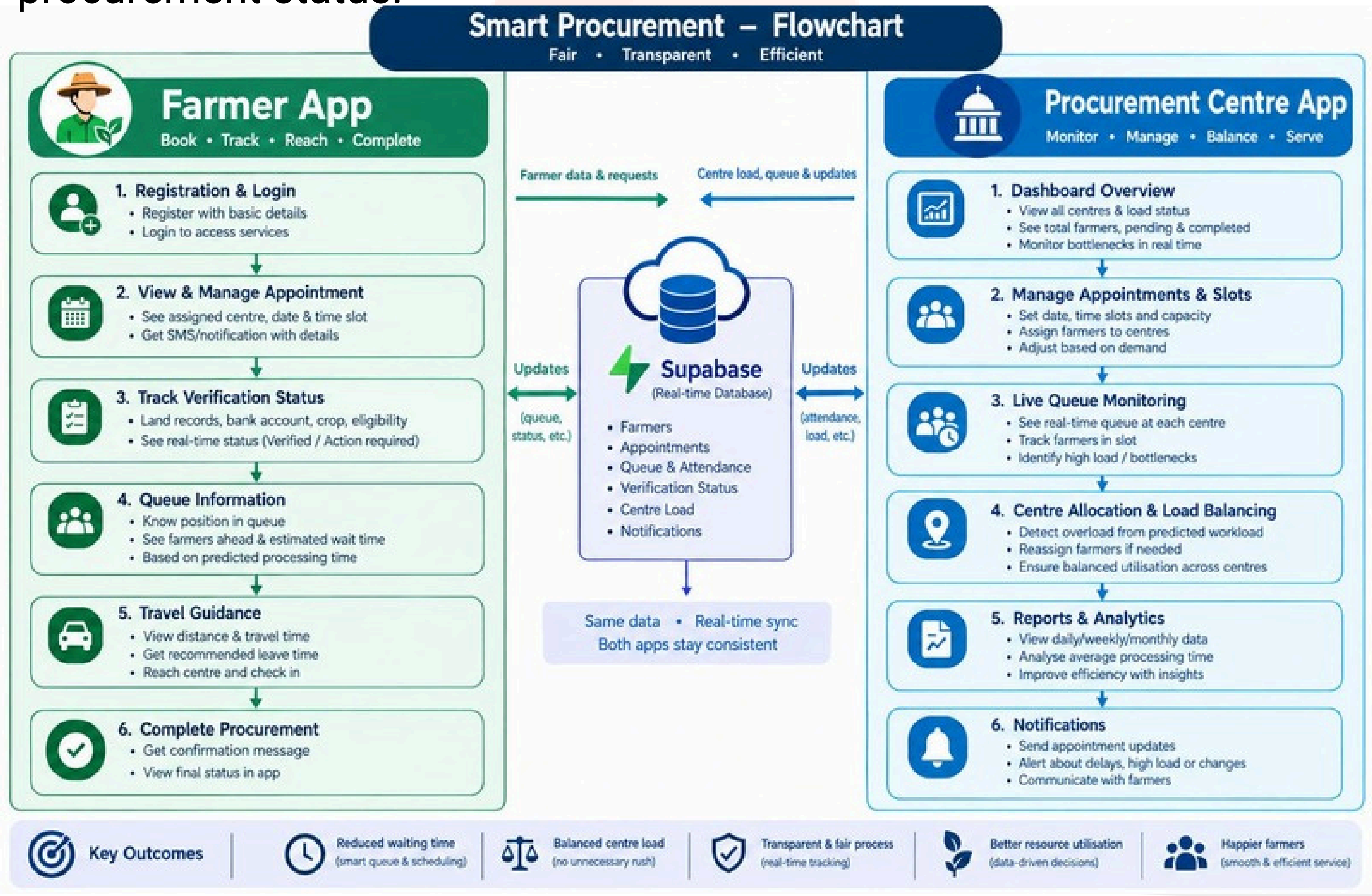
- Problem Statement ID – 26032
- Problem Statement Title- Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainity about procurement status.
- Theme- Smart automation
- PS Category- Software
- Team ID-
- Team Name - Bug Slaying Developer Crew

# Proposed Solution

- **Farmer Registration & Slot Booking:** Farmers Register through AADHAR CARD and request for slots.
- **Real-Time Queue Management:** Backend tracks centre capacity, farmer attendance, and queue position to show farmer live track of procurement centre.
- **Smart Attendance & Travel Tracking:** We take attendance of farmer whether they left the home for procurement centre or not, for rescheduling in emergency cricis and bottlenecks of procurement centre.
- **Dynamic Rescheduling:** If any procurement centre getting bottleneck due to any reason then our app reschedule or redirect farmers who doesn't left from home to nearby procurement centre to use all centres very efficiently and reducing congestion at the procurement centres.
- **Instant Notifications:** Android notifications / SMS-ready architecture informs farmers about slot confirmation, rescheduling and attendance and there is tracking of process in farmer app to track the whole procurement process.
- **Making/assigning of slots:** Our model itself assigns the day and slots to farmers according to the predetermined priority order, which improves the odds of the farmers, who have more urgency/requirement getting an earlier slot.

## Smart Farmer Procurement Management System

**Problem:** Farmers often face long waiting times, lack of information regarding procurement schedules and uncertainty about procurement status.



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# TECHNICAL APPROACH

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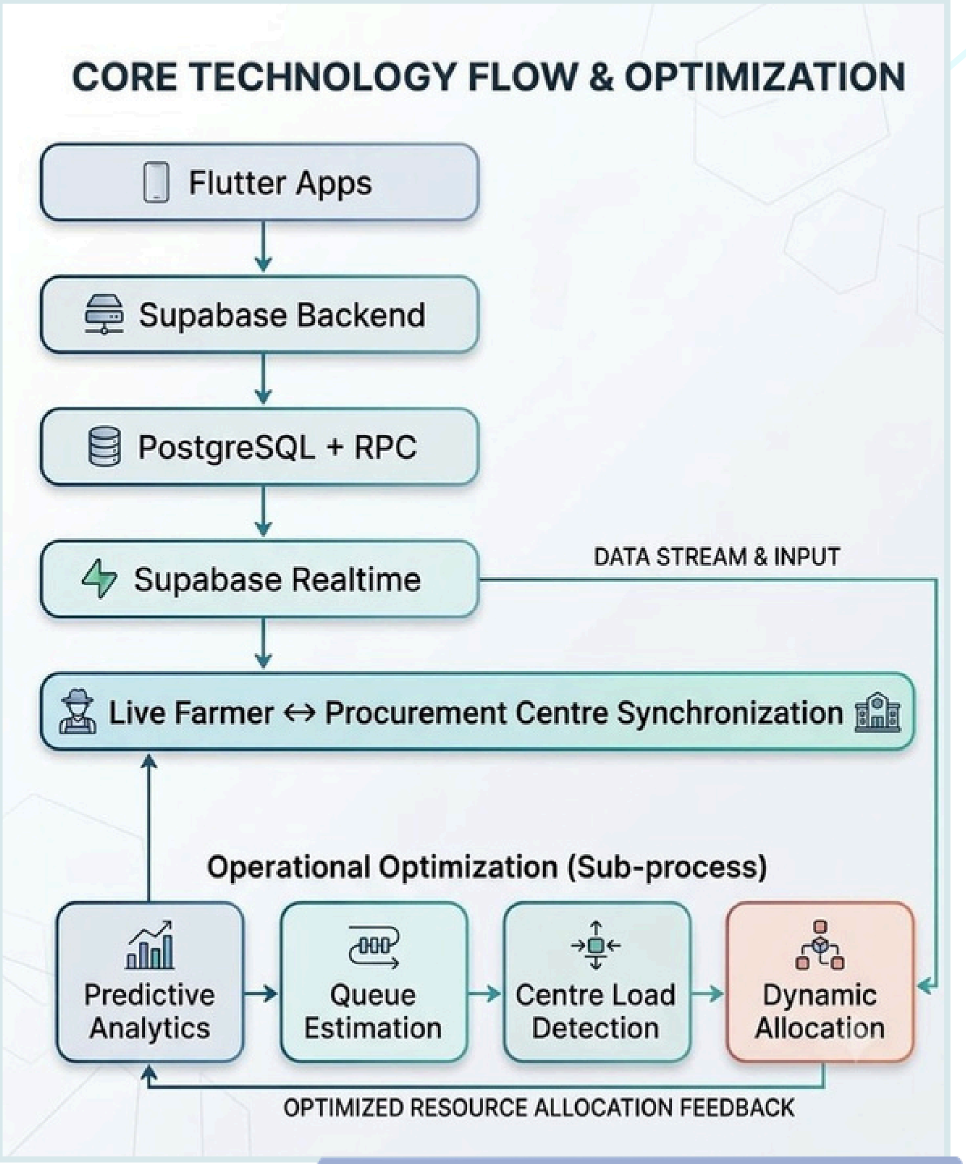
| Category | Technology / Tool | Purpose |
|---|---|---|
| Mobile Frontend | Flutter + Dart | Used to make farmer and procurement centre app. |
| Backend | Supabase | Used for cloud backend and API layer. |
| Database | PostgreSQL | Relational database storing all data (Centralized procurement data). |
| Real-Time Sync | Supabase Realtime | Enables live synchronisation between Farmer and Procurement centre app. |
| Business Logic | PostgreSQL RPC + PL/pgSQL | Used for atomic workflow operation and state management such as attendance. |
| Notifications | Flutter Local Notifications | Used to provide OTP, appointment & status alerts. |
| AI / Analytics | Predictive processing-time model + Queue/Capacity Analysis | Used to provide Estimated Time of Arrival, load detection & dynamic scheduling. |




## Maps & Navigation

- OpenStreetMap (OSM) — Map data.
- flutter_map + LatLong2 — Map rendering and geographic coordinates.
- OSRM (Open Source Routing Machine) — Road-based route, distance and travel-time calculation.

## Languages used

- PL/pgSQL
- Dart
- Python
- Java

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# FEASIBILITY AND VIABILITY

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## Feasibility

### Operational Feasibility

- **Dynamic Scheduling of Slots:** Allowing centres to use their resources very efficiently and never let any procurement centre overcrowded.
- **Dynamic Rerouting:** Automatically redirecting farmers who didn't left the house for procurement centre making easy for everyone. Farmers who are lined up at procurement centres can be accommodated in place of redirected and rescheduled farmers. Redirected and rescheduled farmers will get unhurdeled procurement process.
- **Integration with government:** It can be easily connected to government databases using APIs.

### Technical Feasibility

- **Unique registration:** Only aadhar card holders are eligible to make it secure real and unique registration.
- **Server stability:** To handle thousands of farmers requesting slots at the exact same time without the database crashing, we are using Supabase, PostgreSQL and Realtime to scale. This ensures high-speed, concurrent read/write operations so that two farmers are never double-booked for the exact same time slot.
- **Surviving rural internet blackouts:** For critical state changes—like checking the "attendance" box or executing emergency rerouting — the system relies on an SMS-gateway fallback. If a network drops, the database state is synchronized via standard text messages, keeping the queue alive under any conditions.

### Infrastructure & Hardware Feasibility

- **Low Compute Architecture:** The core logic matching farmers to available slots does not require heavy processing power.

## Viability

### Operational Viability

- **Dynamic Scheduling:** Our backend distribute farmers based on actual capacity of centers and predicted workload to prevent overcrowding and inefficient working of procurement centres.
- **Automated Crisis Rerouting:** Equipment breaks down and delays happen. The inclusion of the redirect feature ensures that a single broken machine does not cause a massive, days-long traffic jam. Automated load-balancing keeps the operational flow moving, which is vital for real-world survival.

### Social & Economic Viability

- **DRestoring Farmer Trust:** The biggest barrier to digital adoption in agriculture is a lack of trust. Our transparent tracker provides absolute clarity on crop quality, billing, and payment status, which directly incentivizes farmers to use the platform.
- **Universal Accessibility via SMS:** A digital solution fails if the end-user cannot access it. By utilizing offline SMS fallbacks for critical updates like slot allotment or redirect alerts our system remains highly usable and viable for farmers in remote areas with poor internet connectivity.
- **Low-Cost Ecosystem Integration:** Building a massive, standalone database from scratch is expensive and hard to maintain. By integrating with existing digital public infrastructure (like Aadhaar for secure), and using cloud based deployment makes our system incredibly cost-effective and easy for the government to scale nationally.

## Challenges

**Low Digital Literacy Among Farmers**
Many farmers may not be familiar with mobile apps or online slot booking.
**Solution:** Provide a simple and multilingual app interface and increasing awareness towards farmers for the app.

**High Number of Users During Peak Procurement Season**
A large number of farmers may use the system simultaneously.
**Solution:** Use scalable cloud infrastructure and efficient databases to handle high traffic and prevent system failure.

**Network Connectivity limitations in rural areas**
There are some rural areas with limited internet connectivity.
**Solution:** The architecture can be extended with local caching, retry mechanism and synchronisation when connectivity is restored.

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# IMPACT AND BENEFITS

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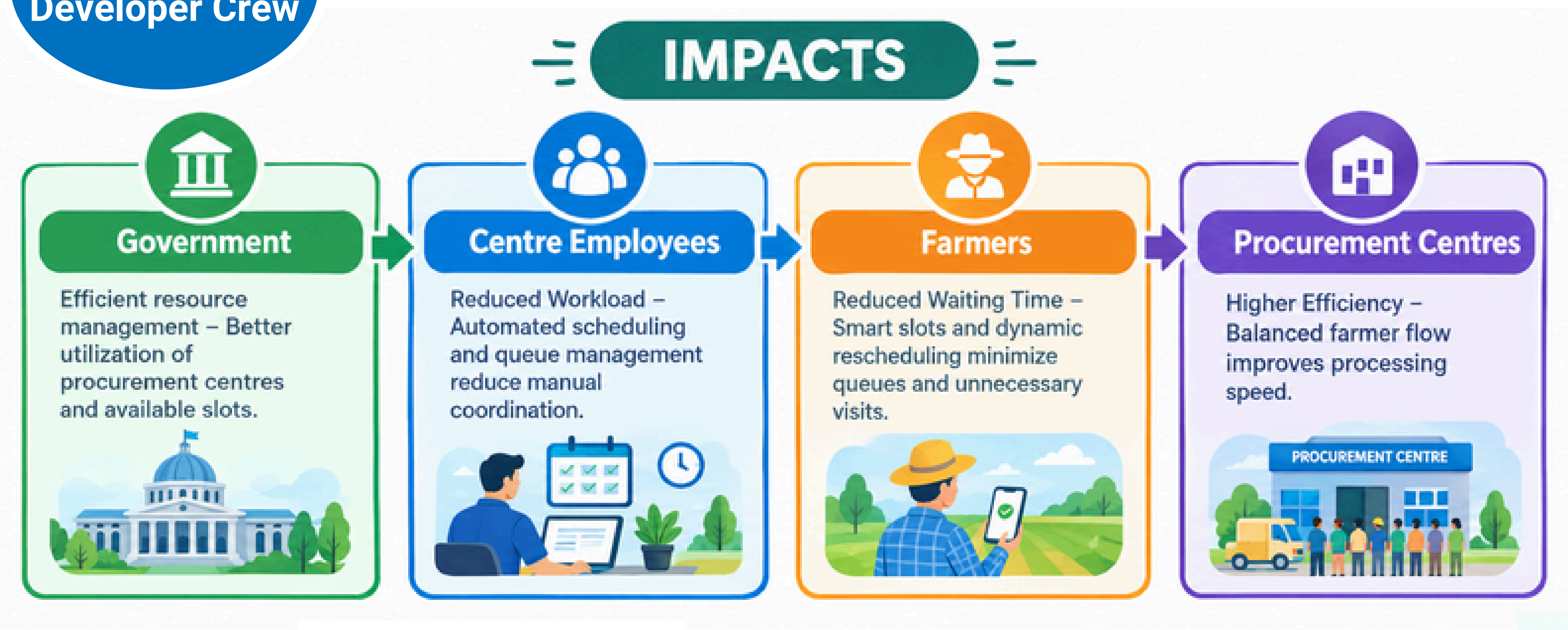



## BENEFITS

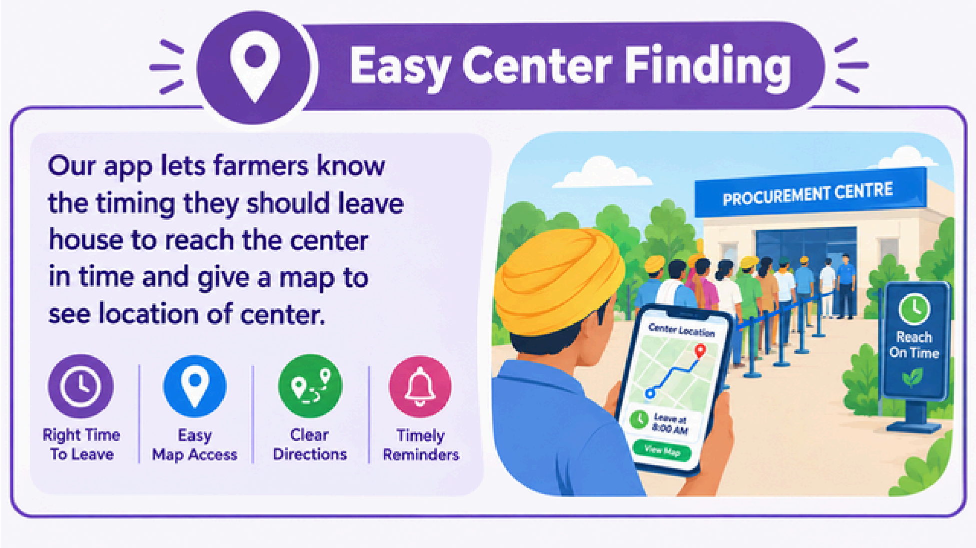



Increased Trust

Easy to use

No queue

Resource Efficiency



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# RESEARCH AND REFERENCES

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## Researched from these websites

- https://enam.gov.in
- https://agmarknet.gov.in
- https://epaddy.wb.gov.in
- https://ekharid.haryana.gov.in/Search.aspx
- https://mpeuparjan.mp.gov.in/mpeuparjan25/Home.aspx
- https://hpappp.hp.gov.in/

- **Google Drive Link for APKs:** https://drive.google.com/drive/folders/1WwD6o_8Cqlro9Vq1l0U77QsJfokM6BGC?usp=sharing
- **Demo Video Link:** https://drive.google.com/drive/folders/1g_aWKB6h5VGu0_oUmFHOLTpofaUu8L0i?usp=sharing
- **Git-Hub Repository:** https://github.com/CodeAbhiGYAN/SIH-AGRIPROCURE-2026